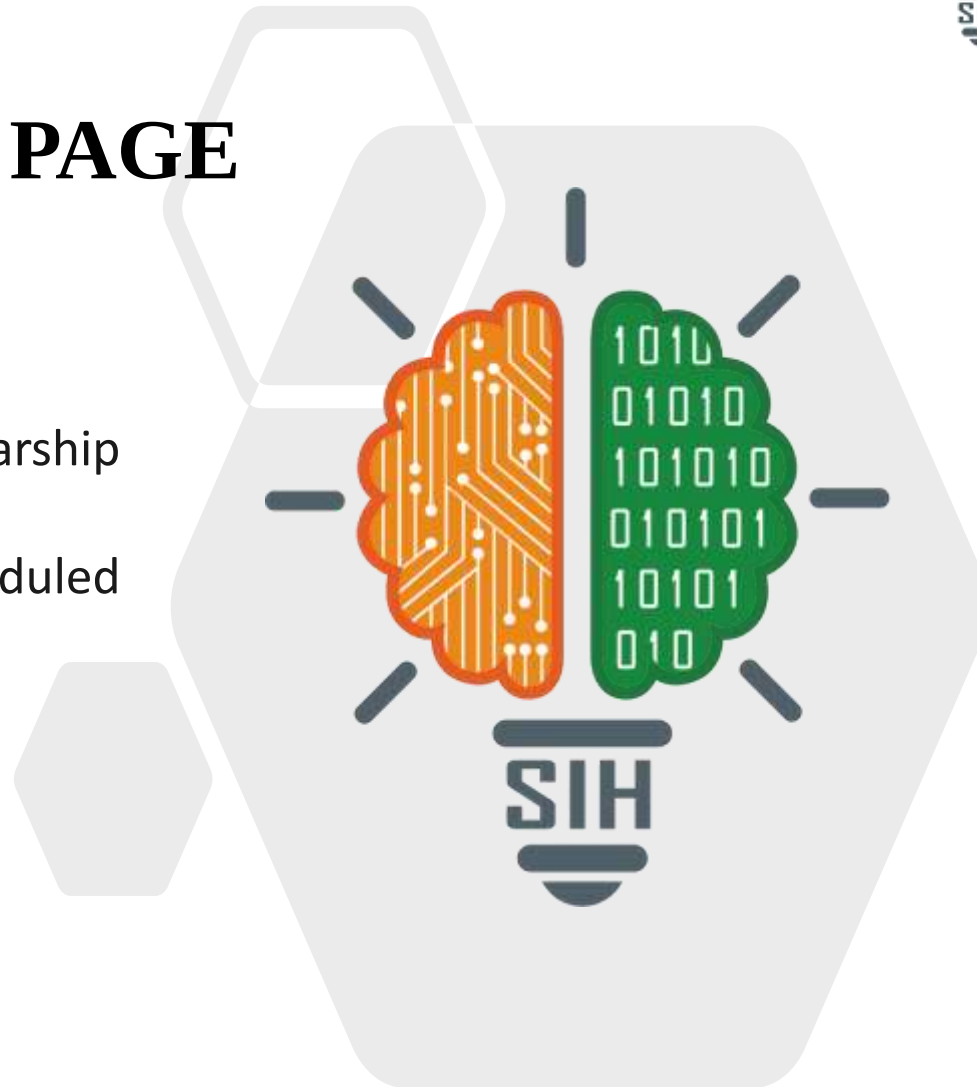


TITLE PAGE

- **Problem Statement ID** – 26239
- **Problem Statement Title-** AI-Enabled Scholarship and Fellowship Management System for Scheduled Tribes
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-** 154583
- **Team Name-** SMRITII



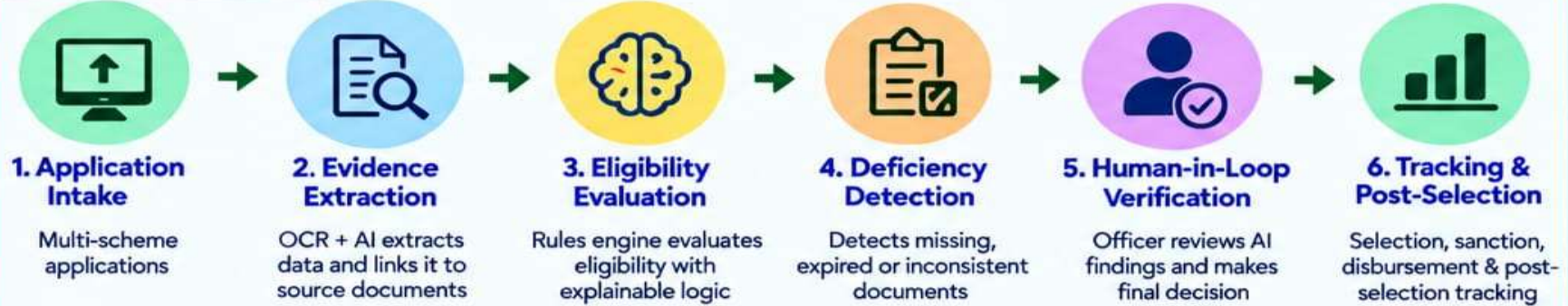
An AI-powered ,configurable scholarship administration platform that transforms fragmented applications and documents into an explainable evidence –driven workflow for the Ministry of Tribal Affairs.

THE PROBLEM

- Fragmented applications & documents across schemes.
- Manual verification causes delays, errors and rejections.
- Inconsistent rules and workflows.
- Lack of transparency and audit trail.



HOW IT WORKS



OUR SOLUTION

A configurable, AI-powered platform that automates verification, detects deficiencies and streamlines end-to-end scholarship administration across schemes.



CONFIGURABLE SCHEME ENGINE



EXPLAINABLE DECISIONS

✓ Why Eligible?	Rule: Age 18–35 Evidence: DOB in Aadhaar.pdf
✗ Why Deficient?	Missing Income Certificate Required as per Rule 2.3
i What Changed?	Income exceeds limit Limit: ₹2,50,000 Found: ₹3,10,000

CORE INNOVATION



IMPACT

- ✓ Faster scrutiny
- ✓ Fewer avoidable rejections
- ✓ Transparent decision-making
- ✓ Reduced administrative workload
- ✓ Reusable for multiple tribal schemes

Frontend

React 18 + TypeScript, Vite, Tailwind CSS, React Router, Axios, Recharts

Backend

Java 21 + Spring Boot — Spring Security, JWT, Spring Data JPA/Hibernate, Bean Validation, OpenAPI/Swagger

AI / Verification

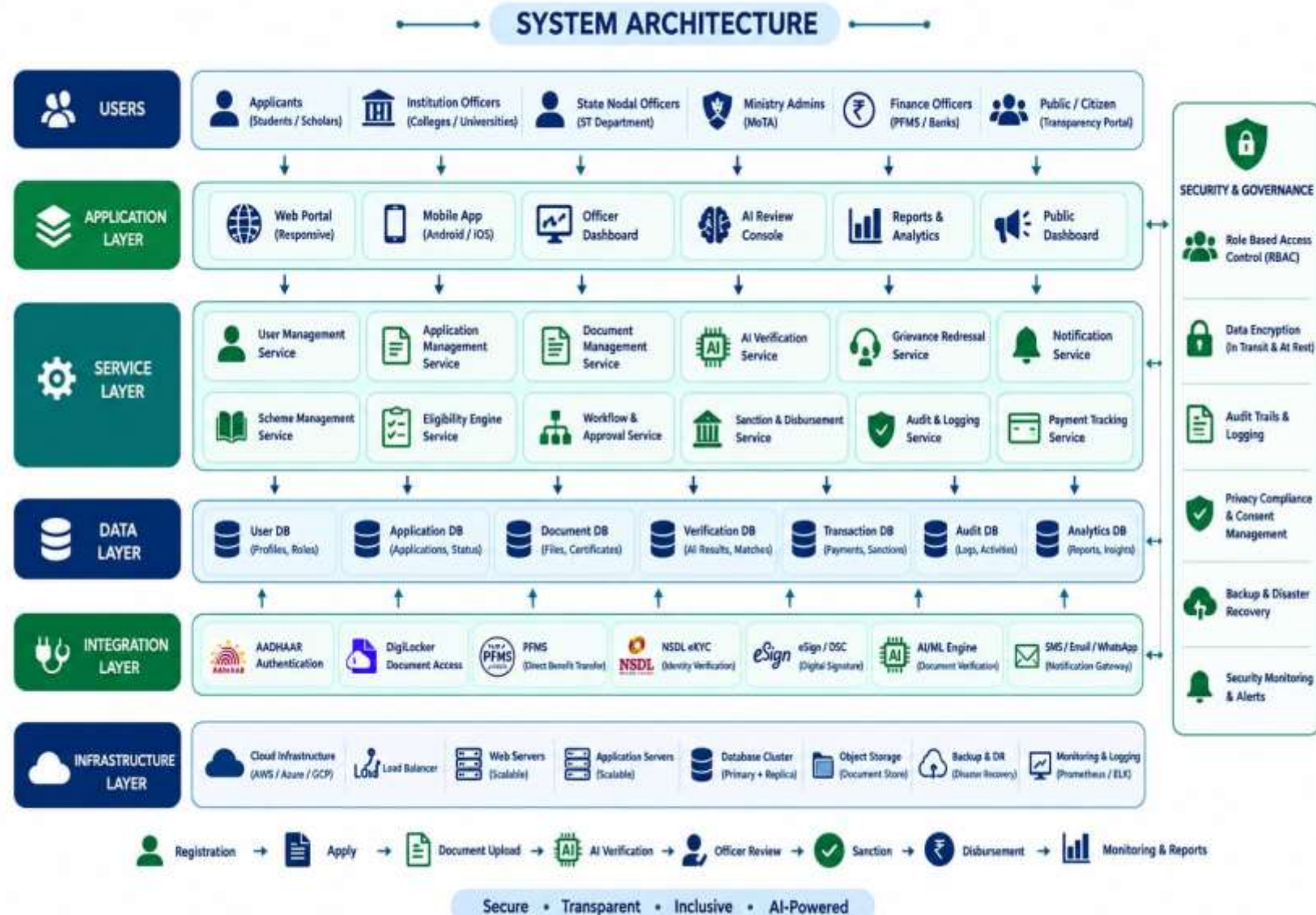
Python + FastAPI — OCR (Tesseract), Apache Tika/PDFBox, NLP field extraction, consistency & anomaly checks

Data Layer

PostgreSQL (system of record) · Redis (cache/session) · S3-compatible object storage (documents)

Events & DevOps

Apache Kafka (async events) · Docker + GitHub Actions CI/CD
GCP (Cloud Run, Cloud SQL, Cloud Storage)



TECHNICAL

- Built with **React, Spring Boot, FastAPI & PostgreSQL**
- **Modular + REST API** architecture
- OCR + rule-based document validation
- Human-in-the-loop AI for safe verification
- Docker-based, scalable deployment

OPERATIONAL

- End-to-end workflow: **Application → Verification → Deficiency → Approval → Disbursement**
- Role-based dashboards for **students & officials**
- Automated notifications reduce manual follow-ups
- **Audit logs** ensure traceability

IMPLEMENTATION

- Deploy in phases: **Application → Documents → Verification → Sanction → Tracking**
- Cloud-ready for increasing application volume

Financial

- Open-source technologies → lower licensing cost
- Automation → reduced manual workload
- Modular design → no complete rebuild for new schemes

Social

- Easy scholarship discovery & application tracking
- Clear deficiency/correction notifications
- Mobile-friendly & simple workflows

Institutional

- One platform for multiple schemes
- Role-based access protects applicant data
- Analytics identify bottlenecks & scheme-level trends

“SAARTHI is a reusable digital infrastructure for multiple scholarship & fellowship schemes — not just a one-time portal.”

Target audience: ST scholarship/fellowship applicants, verifying officers, scheme administrators, and the Ministry of Tribal Affairs.

Social

- **Easier access** to scholarship & fellowship opportunities
- **Fewer avoidable rejections** through deficiency correction
- Transparent **application status tracking**
- Faster communication between **applicants & authorities**
- Supports users with **different levels of digital literacy**

Economic / Operational

- Reduces **manual verification workload**
- Lowers administrative & operational costs
- **Open-source + cloud-ready** architecture reduces infrastructure cost
- Faster processing can reduce delays in **sanction & disbursement**
- Reusable platform avoids rebuilding systems for every scheme

Governance

- **One platform for multiple schemes**
- Explainable eligibility and deficiency decisions
- **Audit trail** for important actions
- Real-time dashboards for **SLA & bottleneck monitoring**
- Data-driven insights for **better scheme administration**

Official Government Sources

Ministry of Tribal Affairs — Scholarship Schemes — tribal.nic.in/ScholarshiP.aspx

Scheme guidelines for Pre-Matric, Post-Matric, Top Class Education, National Fellowship & National Overseas Scholarship.

DBT Tribal — Active Scheme Catalogue — dbttribal.gov.in/AllScheme.aspx

Live list of Direct Benefit Transfer schemes for ST students used as architectural/data-model grounding.

Ministry of Tribal Affairs — Knowledge Hub — tribal.nic.in/KnowledgeHub.aspx

Scheme guidelines, amendments and FAQs referenced for eligibility-rule templates.

Our Prototype :

[Live Demo](#)